MAHATMA GANDHI UNIVERSITY KOTTAYAM

MGU-BVOC(HONOURS) REGULAR EXAMINATION APRIL 2026

SECOND SEMESTER

MPC-MG2MPCCHY100-BASIC ORGANIC CHEMISTRY

(2025 ADMISSION ONWARDS)

Part A: Answer in Two Sentences

Each question carries 2 marks.

1. Compare the geometry of a carbocation versus a carbanion.
   A carbocation typically has a trigonal planar geometry with sp^2 hybridization and a bond angle of approximately 120°. In contrast, a carbanion generally exhibits a trigonal pyramidal geometry with sp3 hybridization, similar to ammonia.
2. Describe the characteristic shape of an sp hybridised carbon atom.
   An sp hybridized carbon atom has a linear shape. The two sp hybrid orbitals are oriented at a bond angle of 180° to each other.
3. Give an example for Anti-Markownikoff addition.
   A classic example is the hydroboration-oxidation of propene or the addition of HBr to propene in the presence of peroxides (Kharasch effect), which yields 1-bromopropane.
4. Identify the electrophile involved in the sulphonation of benzene.
   The active electrophile in the sulphonation of benzene is sulfur trioxide (SO3). Even though it is neutral, the sulfur atom is highly electron-deficient.
5. Identify the specific reactive intermediate formed during the electrophilic addition of HBr to ethene.
   The reactive intermediate formed is a carbocation, specifically the ethyl carbocation ($CH3CH2^{+}$).
6. Define the term Enantiomer.
   Enantiomers are a pair of stereoisomers that are non-superimposable mirror images of each other. They have identical physical properties except for the direction in which they rotate plane-polarized light.
7. State Auwers-Skita rule.
   The Auwers-Skita rule states that among alicyclic diastereomers, the isomer with the higher boiling point, higher refractive index, and higher density is generally the cis-isomer.
8. Draw the most stable and unstable conformation of cyclohexane.
   The chair conformation is the most stable due to minimized torsional and steric strain. The half-chair is considered the most unstable, though the boat conformation is also a high-energy, unstable form.
9. Identify the configuration if the two groups of lowest priority are on the same side of the double bond.
   In the E/Z system, if the two higher-priority groups are on the same side, it is the Z-configuration (*zusammen*); if the question implies the "lowest" priority groups are on the same side (meaning high priority are also on the same side), it is Z.
10. List two examples of synthetic rubber.

Two common examples are Neoprene (polychloroprene) and Buna-S (Styrene-Butadiene Rubber).

11. Illustrate the simple repeating pattern of a block copolymer using A and B notations.
    A block copolymer consists of long sequences of one monomer followed by long sequences of another, represented as: -A-A-A-A-B-B-B-B-.
12. Recall the repeating unit of Polyethylene.
    The repeating unit of polyethylene is -(CH2-CH2)-.

Part B: Short Answer

Each question carries 5 marks.

13. Write a note on electrophiles.
    Electrophiles are electron-loving species that are either positively charged (H^+, $NO2^+$) or neutral molecules with vacant orbitals (BF3, AlCl3). They attack regions of high electron density during a chemical reaction.

14. Write a note on homolytic and heterolytic fission.

- Homolytic Fission: A covalent bond breaks such that each atom retains one electron from the shared pair, resulting in the formation of free radicals.
- Heterolytic Fission: The bond breaks such that both electrons are retained by one of the atoms (usually the more electronegative one), resulting in the formation of ions (cations and anions).

15. Compare E1 and E2 reactions.

- E1 (Unimolecular Elimination): Occurs in two steps via a carbocation intermediate; the rate depends only on the substrate concentration.
- E2 (Bimolecular Elimination): Occurs in a single concerted step where the base abstracts a proton at the same time the leaving group departs; the rate depends on both the substrate and the base.

16. Compare polar and non-polar molecules.

- Polar Molecules: Have an uneven distribution of electron density due to differences in electronegativity between atoms, resulting in a permanent dipole moment (e.g., H2O).
- Non-polar Molecules: Electrons are shared equally (e.g., Cl2) or the molecular geometry causes bond dipoles to cancel out (e.g., CO2), resulting in a zero net dipole moment.

17. Explain why a Meso compound is optically inactive.

A meso compound contains chiral centers but is optically inactive because it possesses an internal plane of symmetry. The optical rotation caused by one half of the molecule is exactly cancelled by the equal and opposite rotation of the other half (internal compensation).

18. Identify and explain the two types of optical rotation.

- Dextrorotatory (d or +): A compound that rotates the plane of polarized light to the right (clockwise).
- Laevorotatory (l or -): A compound that rotates the plane of polarized light to the left (counter-clockwise).

19. Describe the step-by-step preparation of Nylon 6,6. Nylon 6,6 is prepared by the condensation polymerization of adipic acid and hexamethylenediamine.

1. Equal molar amounts of the two monomers are reacted to form "Nylon Salt."
2. The salt is heated under vacuum to eliminate water, linking the monomers via amide bonds to form the long-chain polymer.

20. Classify polymers based on intermolecular forces.

- Elastomers: Very weak forces (e.g., Buna-S).
- Fibers: Strong intermolecular forces like hydrogen bonding (e.g., Nylon 6,6).
- Thermoplastics: Intermediate forces; can be softened by heating (e.g., Polyethylene).
- Thermosetting Polymers: Heavily cross-linked; cannot be remolded (e.g., Bakelite).

Part C: Essay

Each question carries 10 marks.

21. Write an essay on Inductive effect.

The inductive effect is the permanent displacement of sigma (sigma) electrons along a chain of carbon atoms due to the presence of an atom or group of different electronegativity.

- -I Effect (Electron-withdrawing): Groups like -NO2, -Cl, and -COOH pull electrons away from the carbon chain.
- +I Effect (Electron-releasing): Groups like alkyl groups (-CH3) push electrons toward the carbon chain. It is a distance-dependent effect that weakens as it moves further from the substituent.

22. Write an essay on SN1 and SN2 reactions.

- SN1 (Unimolecular Nucleophilic Substitution): A two-step mechanism involving a carbocation intermediate. It is favoured by tertiary halides and polar protic solvents. The reaction often results in racemization.
- SN2 (Bimolecular Nucleophilic Substitution): A one-step concerted mechanism involving a transition state. The nucleophile attacks from the backside, leading to complete inversion of configuration (Walden Inversion). It is favoured by primary halides.

23. Describe in detail the conformational analysis of n-butane. Rotation around the C2-C3 bond of n-butane results in several conformations:

- Fully Eclipsed: Highest energy due to maximum steric and torsional strain.
- Gauche: Staggered but with methyl groups close to each other (intermediate stability).

- Eclipsed: High energy.
- Anti (Staggered): Lowest energy and most stable because the bulky methyl groups are 180 degrees apart.

24. Compare and contrast addition and condensation polymerization.

- Addition Polymerization: Monomers (usually containing double bonds) join together without the loss of any small molecules. The polymer has the same empirical formula as the monomer (e.g., Polyvinyl Chloride).
- Condensation Polymerization: Monomers (containing bifunctional groups) react with the elimination of small molecules like H_2O, NH_3, or HCl. The process is step-wise (e.g., Polyesters, Terylene).